CAB4155644.1
LR796631.1:3228-22784
uncultured Caudovirales phage

Phage_capsid_2 ? E1_FCCH CBM_5_12 ? T7_gp14 ? ART-PolyVal PBECR5 ? ? ? Gly_rich_2 ? ? ?

CAB5220657.1
LR798286.1:4120-25469

uncultured Caudovirales phage

Phage_capsid_2 ? ? ? ? ? SLT LPD22 Gly-zipper_YMGG PBECR5 Pect-lyase_RHGA_epim ? ? Glyco_hydro_108 ? Arm-DNA-bind_3 Phage_int_M Phage_integrase ?

CAL62400.1
CU207211.1:2256939-2277616
Herminiimonas arsenicoxydans

SU10_adaptor ? ? ? ? ? ? PBECR5 LPD38 Phage_lysozyme ? HSDR_N_2 Phos_pyr_kin EAL GGDEF dCache_1 ? ?

CCX39506.1
HF985250.1:412-18514
Firmicutes bacterium CAG:102

? ? Phage_XkdX ? ? Gp58 Peptidase_S74 ? LPD3 PBECR5 ? ? ? Amidase_2 DUF4406 DUF6275 Mn_catalase

CCY12070.1
HF988773.1:595-21903
Clostridium sp. CAG:81

Phage_XkdX Sequestin ? ? Peptidase_S74 ? DUF3597 PBECR5 GAS ? Phage_lysozyme Choline_bind_3 DcuC Peptidase_M28 Peptidase_M20 CitMHS

DAD87026.1
BK015012.1:3324-20705
Myoviridae sp. ctRRy11

RVT_1 ? LRRs DivIC DUF6057 ? CFSR ART-PolyVal PBECR5 Zn_ribbon_recom Recombinase Resolvase ? HTH_3 HTH_3 ? ? IL7

DAD90516.1
BK015088.1:6792-23406
Myoviridae sp. ctiBE32

? ? ? ? ? ? ? PBECR3 PBECR5 BppU_N DUF7365 ? Phage_holin_6_1 ELLD ELLD ? HTH_3

DAE36700.1
BK032935.1:1-17213
Caudoviricetes sp.

Phage_XkdX Mtd_N Collagen Phage_holin_4_1 Sequestin Amidase_3 Glucosaminidase NLPC_P60 ? LPD23 DUF6782 MuF_C ART-PolyVal PBECR5 ? Peptidase_S31 DUF4199

DAE37296.1
BK032942.1:1-14489
Bacteriophage sp.

Phage_holin_5_2 ? LPD3 ART-PolyVal PBECR5 ? ? Aromatic_hydrox BMFP DAHL ? ?

DAE47127.1
BK033135.1:25406-47421
Caudoviricetes sp.

Sequestin ? ? ? ? T7_gp14 ? PBECR2 ART-PolyVal LPD3 PBECR5 DNA_methylase ART-PolyVal LPD38 ? ? ? ? ? NUMOD4 HNH_3 FAM81B

DAE74078.1
BK019634.1:13338-26140
Caudoviricetes sp.

? ? Phage_capsid_4 ? ? AAA_8 ? PBECR5

DAE81671.1
BK019768.1:13919-34433
Caudoviricetes sp.

? GP7 Bbp16 ? ? T7_gp14 ? ddrB-ParB PBECR5 ? Collagen Collagen Collagen Phage_tail_APC ? ? ?

DAF49727.1
BK032586.1:20785-39835
Podoviridae sp. ctZih56

Collagen Collagen Collagen Collagen Collagen Collagen Collagen Big ? CBM_5_12 ? ? ? MuF_C PBECR5 Phage_holin_6_1 ? ? Sigma70_r4 ? PfaD_N ?

DAF61464.1
BK032812.1:1278-19126
Podoviridae sp. ctzXp5

Arm-DNA-bind_5 Phage_int_SAM_5 ? DUF6965 ? ? ? ? InPase PBECR5 ? ? ? ? ? ? ?

DAF99364.1
BK016164.1:23433-43135
Myoviridae sp. ct7113

Amidase_2 Gp58 Peptidase_S74 ? ? ? LPD23 PBECR5 UPF0547 ? ? HTH_38 rve HTH_38 ? Hydrolase_2

DAG21522.1
BK017667.1:1-10144
Caudoviricetes sp.

MuF_C PBECR5 ? ? ? ? ? ? Phage_r1t_holin

DAG26479.1
BK017747.1:4639-22905
Caudoviricetes sp.

? Phage_holin_3_3 Peptidase_M15_3 Sequestin ? ? ? LPD3 PBECR5 ddrB-ParB ? ? ? ? ? ? Phage_integrase Phage_int_M Arm-DNA-bind_3

DAG31603.1
BK017842.1:3844-24417
Caudoviricetes sp.

Phage_capsid_4 ? ? ? ? ? ? PBECR5 ? ? ? DUF6273 ? ? RVT_1

DAG46590.1
BK018117.1:17268-39596
Caudoviricetes sp.

? ? ? ? ? DUF859 DUF859 Gp10_C ? PBECR5 ? ? CBM_5_12 Phage_holin_5_2 Amidase_2 PG_binding_1 HTH_24 ?

DAG70164.1
BK016299.1:24320-47075
Caudoviricetes sp.

Phage_int_SAM_3 Phage_integrase ? GP3_package Terminase_3 ? ? Flg_new PBECR5 ? ? ? ? ? ? LRRs

DAG77537.1
BK016475.1:1-11717
Caudoviricetes sp.

? ? MuF_C PBECR5 Phage_T7_tail ? ? ? ? ? ?

DAG91468.1
BK035291.1:11776-38989
Herelleviridae sp.

? ? ? ? ? ? ? LPD3 PBECR5 Big Big Big DUF6273 ? Calcineurin Phage_integrase Peptidase_M15_3 Phage_holin_4_1 Holin_SPP1

DAH33873.1
BK018996.1:7166-27423
Caudoviricetes sp.

? ? ? Gly_rich_2 Phage_T7_Gp13 T7_gp14 NLPC_P60 LPD3 MuF_C PBECR5 Phage_T7_tail ? ? ? ? ? ?

DAH82214.1
BK024233.1:10213-26467
Caudoviricetes sp.

? ? ? ? ? ? ? PBECR5 MuF_C ART-PolyVal Phage_integrase ? ? DUF7841 ? BclA_C ?

DAI02728.1
BK024655.1:2551-18605
Caudoviricetes sp.

? Phage_holin_Dp1 CHAP PG_binding_1 ? ? CCDC39 ? PBECR5 Sigma70_r4 ? ? ? ? ? Antimicrobial_8

DAI99966.1
BK020800.1:7588-22549
Caudoviricetes sp.

? ? ? ? ? ? ? PBECR1 ddrB-ParB InPase ART-PolyVal PBECR5 LPD38 HTH_3 HigB_toxin ?

DAJ43157.1
BK043284.1:17409-35663
Caudoviricetes sp.

? Phage_capsid_4 ? ? ? Gp10_C ? PBECR5 ? ? ? Big ? ? ?

DAJ80279.1
BK049754.1:26066-44801
Caudoviricetes sp.

? ? ? ? PBECR5 ? ? ? ? ? DUF7841 ?

DAJ90406.1
BK049956.1:15388-35151
Caudoviricetes sp.

Phage_capsid_3 ? ? ? Peptidase_S74 ? ? ? Collagen ? ? TssO ? CHAP ?

DAK10132.1
BK035475.1:10290-29676
Bacteriophage sp.

DUF2798 ? Amidase_2 Gp58 Peptidase_S74 ? ? ? LPD23 MuF_C PBECR5 ? ? ? ? ? tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2

DAK79107.1
BK047805.1:7174-23844
Caudoviricetes sp.

? ? ? SSB ? ? ? PBECR5 ? ? ? ? SCD2_helical_N Rep3_C Rep3_N ?

DAK92788.1
BK048074.1:38647-53912
Caudoviricetes sp.

PBECR5 HNH ? ? Tad8 DUF551 DUF2197 ?

DAL32411.1
BK031799.1:23492-42903
Caudoviricetes sp.

? ? ? ? ? ? ? PBECR5 ? Big Big Big Big ? Big

DAM94132.1
BK052955.1:2943-23611
Caudoviricetes sp.

Phage_capsid_3 ? ? ? ? ? ? LPD23 PBECR5 ? ? ? ? ? RVT_1 ?

DAO77241.1
BK038095.1:14545-32058
Bacteriophage sp.

? Phage_holin_4_1 NAGPA Peptidase_M15_3 ? ? CCDC39 ? PBECR5 ? ? ? ? BclA_C DUF7841 ?

DAP05494.1
BK043925.1:25-20477
Caudoviricetes sp.

? ? Amidase_2 Gp58 ? ? ? LPD23 LPD3 PBECR5 ART-PolyVal ? HTH_3 HTH_38 rve

DAP29646.1
BK044407.1:2275-19951
Caudoviricetes sp.

? ? ? HicA_toxin UBA_DHX29 DUF3761 CAP ? ApoLp-III ? ? ? Collagen Collagen ? DUF1664 Amidase_5 ZoocinA_TRD

DAP45359.1
BK044686.1:23705-43629
Caudoviricetes sp.

CCDC34 Phage_capsid_4 ? ? ? ? PG_binding_1 MuF_C PBECR5 BppU_N Collagen Collagen

DAP62541.1
BK045038.1:23290-42838
Caudoviricetes sp.

DUF6275 ? ? GP3_package ? SU10_portal ? Peptidase_M78 PBECR3 ART-PolyVal PBECR5 ART-PolyVal ? ? Flg_new ? Zn_ribbon_10 ? ?

DAQ26868.1
BK054094.1:20332-45146
Caudoviricetes sp.

? ? ? ? ? T7_gp14 ? MuF_C PBECR5 P22_portal ? ? ? ? ? ?

DAQ54618.1
BK057964.1:3244-14600
Bacteriophage sp.

Calcineurin ? ? ? ? ? ? PBECR5 Phage_T7_tail ? ? ? ? ? ?

DAQ59707.1
BK058077.1:71-7035
Caudoviricetes sp.

? Phage_scaffold ? ? ? ? Phage_Mu_F PBECR5 ? ? ? ? tRNA-synt_1b tRNA-synt_1b ?

DAQ90343.1
BK058672.1:85-18082
Caudoviricetes sp.

? ? Gp58 Peptidase_S74 ? ? ? ? LPD3 LPD23 SprT-like PBECR5 ? ? DUF4406 Tad2-like Tad2-like ? Amidase_2 TusA Frag1

DAR06209.1
BK058968.1:4004-21083
Caudoviricetes sp.

DUF4376 ABC_trans_CmpB ? ? ? Gp58 LysM PBECR5 ? GAS Tad2-like Phage_r1t_holin Phage_lysozyme PG_binding_1 ? ?

DAR27913.1
BK057106.1:1-19628
Caudoviricetes sp.

DUF859 ? ? Terminase_6N PG_binding_1 ART-PolyVal PBECR5 Ig_VWA7 DUF3378 Amidase_2 PG_binding_1 SH3_3 ? Phage_holin_5_2 Sigma70_r4 ?

DAR38003.1
BK057323.1:8347-28729
Caudoviricetes sp.

? Phage_lysozyme DUF2570 LysC RapA_C ? SU10_portal ddrB-ParB MuF_C ART-PolyVal PBECR5 LPD38 ? ? Phage_capsid_4 ? ? ? ?

DAS20693.1
BK034420.1:271-20269
Caudoviricetes sp.

Bbp16 ? ? ? Phage_T7_Gp13 T7_gp14 ? PBECR5 Pect-lyase_RHGA_epim ? ? ? T6SS_VgrG3-like_C ? ?

DAT56711.1
BK046416.1:7153-27087
Caudoviricetes sp.

? Phage_capsid_3 ? ? ? ? ? LPD3 PBECR5 Collagen ? ? TMP_4 Phage_holin_1 Sequestin DUF6275

DAT88015.1
BK047009.1:5555-26643
Caudoviricetes sp.

Peptidase_Prp ? Phage_capsid_4 ? ? ? ? PBECR5 ? ? ? ? ? DUF6273 ?

DAU40342.1
BK051094.1:4436-22275
Caudoviricetes sp.

? ? ? DUF6275 Phage_holin_5_2 Amidase_2 HopBF1_kinase PBECR5 ART-PolyVal LPD3 GerE HNH ? DUF722 ? Sigma70_r4 Sigma70_r2 ?

DAU60216.1
BK051468.1:4302-27992
Caudoviricetes sp.

P22_portal P22_portal ? Phage_capsid_4 ? ? ? ? PBECR5 Gp58 CBM_4_9 CBM_5_12 ? ? NLPC_P60 PG_binding_1 ? ?

DAU93074.1
BK055053.1:5445-27841
Caudoviricetes sp.

? ? ? ? ? HicB-like_2 HicA_toxin LPD3 PBECR5 Big Big Big Big Big ? ? ? Phage_XkdX Glyco_hydro_25 ?

DAV09770.1
BK055394.1:13336-34218
Caudoviricetes sp.

? DUF4777 ? GP3_package Terminase_6N SU10_portal ? PBECR3 ART-PolyVal PBECR5 ART-PolyVal Zn_ribbon_10 ? ? ? ? Phage_capsid_4 ?

DAV27154.1
BK055765.1:23218-39024
Caudoviricetes sp.

? ? ? ? ? ? ? MuF_C PBECR5

DAW03102.1
BK029563.1:814-26469
Caudoviricetes sp.

Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase HTH_26 ? HTH_SUN2 ? T7_gp14 ? ddrB-ParB InPase RadC PBECR5 LPD38 P22_portal ? ? ? ? ? ?

DAW33960.1
BK036425.1:1473-24295
Bacteriophage sp.

Transposase_20 DEDD_Tnp_IS110 ? ? ? DUF859 ? PG_binding_1 PBECR5 LPD3 ? Phage_r1t_holin ? ? ? ? ?

DAX06486.1
BK027374.1:1-21070
Caudoviricetes sp.

? LPD14 PBECR2 PBECR2 LPD3 PBECR5 MuF_C MuF_C P22_portal N6_N4_Mtase SNase ? RE_MjaI T1RH-like_C ? N6_N4_Mtase

DAY12593.1
BK040929.1:13311-37638
Caudoviricetes sp.

? ? ? Phage_holin_4_1 VanY Peptidase_M15_4 T6SS_VgrG3-like_C CHAP ? MutS_I MTS LPD3 MuF_C PBECR5 LPD38 ? ? Phage_integrase Phage_int_SAM_3 ? Peptidase_M78

DAY76417.1
BK027489.1:18215-38871
Caudoviricetes sp.

Phage_capsid_3 ? ? ? ? ? ? ? BppU_N Collagen ? MTUS1_CCDC69_CC2 Phage_r1t_holin CHAP

EDK12787.1
AAZJ01000027.1:1-6426
Haemophilus influenzae 22.4-21

SU10_portal ? ddrB-ParB MuF_C ART-PolyVal PBECR5

EGR2855325.1
AAXNZZ010000092.1:1-7026
Vibrio parahaemolyticus

? ? NDK Band-7 ? ? ? ?

EXZ06149.1
JGDE01000057.1:354-12596

Bacteroides fragilis str. DS-208

Glyco_hydr_30_2 SusD_RagB SusD-like_3 Plug CarbopepD_reg_2 STN FecR_C FecR DUF4880 Sigma70_r4_2 Sigma70_r2 ? ? ART-PolyVal PBECR5 Pox_F16

HAI60123.1
DMHI01000269.1:359-9869
Xanthomonadaceae bacterium

ddrB-ParB PBECR5 DUF6098 B_solenoid_ydck B_solenoid_ydck B_solenoid_ydck B_solenoid_ydck ?

HAR89213.1
DMVD01000048.1:1-20160
Oscillospiraceae bacterium

T4_RIIA1 ? ? ? Dockerin_1 ? ? LPD23 DUF6782 PBECR5 ?

HAZ61018.1
DNIF01000151.1:2065-21914
Gammaproteobacteria bacterium

GP7 Bbp16 ? ? ? ? SLT LPD22 PBECR5 Collar CAF1A_acidic ? ? TPRs ? ?

HBA35607.1
DNIY01000324.1:1-16522
Gammaproteobacteria bacterium

? ? ? ? ? ? ? LPD22 ART-PolyVal PBECR5 ? ? ? ?

HBC28885.1
DNMR01000115.1:4946-25919
Prevotellaceae bacterium

DevT_C PhageMin_Tail ? ? Plasmodium_HRP ? CotH LRR_EndoS PBECR5 LPD23 DUF6782 MuF_C LPD3

HBJ40131.1
DNXC01000047.1:1-5237
Hyphomonas sp.

SHOCT_2 ART-PolyVal PBECR5 ?

HCT51178.1
DPIT01000051.1:59-3327
Balneola sp.

SET 2OG-FeII_Oxy_3 ? PBECR5 ?

HIR25199.1
DVGS01000120.1:663-8724
Candidatus Egerieimonas faecigallinarum

? ? ? ? DUF3597 PBECR3 PBECR5

HIU77919.1
DVNQ01000014.1:1-10953
Candidatus Avilachnospira avicola

? Peptidase_S74 Peptidase_S74 Peptidase_S74 ? DUF3597 PBECR5

HJD05734.1
DWVI01000004.1:171313-197574
Candidatus Onthoplasma faecipullorum

? ? ? GP3_package Terminase_6N SU10_portal Hint_2 ART-PolyVal PBECR5 COesterase ? Phage_cement_2 ? ? ? Phage_fiber_rpt Phage_tail_C Phage_tail_C bAvd

HKL59302.1
DATFIE010000160.1:1-4701
Sphaerochaeta sp.

? ART-PolyVal PBECR5

HLE33399.1
DATDRF010000229.1:2915-13871
Bacteroidota bacterium

PBECR5 ? ? ? ? ? ? Acb2_Tad1_hairpin

HOR79213.1
DAOSRR010000528.1:1-1131
Anaerolineaceae bacterium

PBECR5

HZJ78823.1
DASRNM010000141.1:22508-33981
Clostridia bacterium

2-oxoacid_dh 2-oxoacid_dh BNR_2 LeuA_dimer IPMS_D2 HMGL-like DUF2229 BcrAD_BadFG BcrAD_BadFG ? ? Phage_scaffold PBECR3 PBECR5 ?

HZJ97562.1
DASRNW010000235.1:1-8839

Oligella sp.

ddrB-ParB Acb1 LPD5 LPD1 PBECR5 ART-PolyVal

KKQ11393.1
LBSG01000087.1:1-6240

Candidatus Moranbacteria bacterium GW2011_GWE1_36_7

PBECR5 ART-PolyVal ART-PolyVal

MAE54580.1
NZAB01000010.1:17501-21747
Porticoccaceae bacterium

? ? ? ? ? HTH_7 ? PBECR5

MAV21950.1
PAES01000020.1:13789-33594
Flavobacteriaceae bacterium

Phage_capsid_2 ? ? ? Acetylase DUF6353 ? PBECR5 ? ? Phage_tail_APC ? ? ? ?

MBA3578918.1
JACDCY010000010.1:641-19996
Gemmatimonadaceae bacterium

Phage_stabilise ? Acetylase ? ? ? ? ART-PolyVal PBECR5 Peptidase_S74 ? ? ? ? ? ?

MBD78236.1
PAVO01000007.1:338954-361252
Crocinitomicaceae bacterium

? ? BRCT HHH_2 HHH_5 DNA_ligase_ZBD OB_DNA_ligase DNA_ligase_aden Nlig-Ia RNB CSD2 OB_RNB ? ? Peptidase_S8 PBECR5 A24_N_bact DNA_gyraseA_C DNA_topoisoIV DNA_gyraseB_C Toprim DNA_gyraseB His-Kinase Receiver CheW ? ? SMC_N

MBD9045305.1
RKER01000018.1:31818-53239
Faecalibacterium prausnitzii

Phage_capsid_4 ? ? ? ? ? ? LPD3 PBECR5 ? DUF6273 DUF3972 ? Glyco_hydro_25 ? Holin_SPP1

MBE6036910.1
SVCU01000009.1:28703-51501
Clostridiales bacterium

PALP 2-oxoacid_dh Peptidase_M20 Sigma70_r2 Sigma70_r4_2 Bps HTH_Rpn_C Yae1_N ? ADPrib_exo_Tox ART-PolyVal ART-PolyVal PBECR5 SSB DUF2130 LrgB LrgA TauE PTA_PTB NMN_transporter

MBE6818245.1
SVOD01000029.1:951-20535
Oscillospiraceae bacterium

? ? GASH ? ART-PolyVal PBECR5 HhH-GPD NUDIX_4 N-OB tRNA-synt_2

MBE6910711.1
SVMM01000025.1:12137-28451
Oscillospiraceae bacterium

GIDA FA_synthesis PP-binding Ribonucleas_3_3 dsrm TPR_PATROL1 HTH_38 ? DNA_methylase MuF_C PBECR5 ? HTH_3 HTH_38 Ribosomal_L33 SecE NusG KOW Ribosomal_L11_N Ribosomal_L11

MBE6993303.1
SVKN01000003.1:28620-46058
Oscillospiraceae bacterium

Cas_Cas5d Cas_Csd1 Cas_Cas7 Cas_Cas4 Cas_Cas1 CRISPR_Cas2 Rotamase_2 LPD3 PBECR5 ? Transglutaminase tRNA_Me_trans tRNA_Me_trans_M tRNA_Me_trans_C EamA MatE MatE Polysacc_deac_1 NRDD

MBF1469774.1
JABZTW010000062.1:1-4662
Prevotella pallens

PBECR5 LPD3 ART-PolyVal

MBM24351.1
PBMB01000002.1:120619-132787
Chloroflexota bacterium

ETRAMP Pet100 ? 7tm_3 MAP17 ? AA_permease_2 PBECR5 MFS_1 ? Ras ? ? ? DUF308

MBO4871896.1
JAFZRH010000008.1:535-10970
Lachnospiraceae bacterium

LPD3 PBECR5 Bps His_Phos_1 HhH-GPD NUDIX_4 DHH DHHA1 RecJ_OB Pribosyltran MRM3-like_sub_bind SpoU_methylase Thiolase_N Thiolase_N Thiolase_C

MBO4978719.1
JAGATX010000033.1:87884-107761
Muribaculaceae bacterium

? Relaxase NikA-like DUF3408 Toprim_2 Zn_ribbon_DnaG AAA_25 DUF3853 PBECR3 ddrB-ParB PBECR5 LPD3 RadC ?

MBO5463791.1
JAGAJB010000038.1:118176-138947
Clostridia bacterium

? Phage_capsid_3 ? ? ? ? ? ART-PolyVal PBECR5 ART-PolyVal KinB_sensor BppU_N Big ? ? ?

MBO5480152.1
JAGAIW010000051.1:1-7502
Clostridia bacterium

? MutS_I ART-PolyVal ART-PolyVal PBECR5

MBO7313547.1
JAGCJB010000276.1:4973-32328
Alistipes sp.

KilA-N NYN HTH_3 ? ? ? ? LPD23 PBECR5 P22_portal ? ? SU10_MCP ? ? ?

MBO8442622.1
JADIMU010000016.1:82305-107559
Candidatus Aphodenecus pullistercoris

? GP7 ? ? ? ? ? ART-PolyVal PBECR5 Fez1 ? ? ? ? RVT_1 ?

MBP3649176.1
JAGBAY010000092.1:127-18035
Clostridia bacterium

DZR_2 ? Phage_capsid_3 ? ? ? ? PBECR5 PG_binding_1 DUF6275 Phage_holin_4_1 Tad2-like

MBP5720425.1
JAGCRA010000149.1:13789-38277
Bacteroidales bacterium

DUF1064 ? ? Terminase_3 P22_portal P22_portal Peptidase_Prp ? PBECR5 ART-PolyVal MuF_C ? Phage_capsid_4 ? ? ? ? ?

MBP5725942.1
JAGCRB010000155.1:135-13623
Clostridia bacterium

? Flg_new_2 Flg_new_2 ? PBECR5

MBP7135993.1
JAGOKA010000032.1:6110-23348
Sphingomonadaceae bacterium

? ? ? ? ? ? Peptidase_M23 LPD22 PBECR5 ART-PolyVal ? ?

MBP7484959.1
JAGODY010000166.1:1-4211
Aquabacterium sp.

PBECR5 ART-PolyVal ART-PolyVal

MBP7740162.1
JAGNKT010000007.1:30971-40324
Brachymonas sp.

LptC ? SMI1_KNR4 adh_short GIY-YIG UvrC_M UVR UvrC_RNaseH UvrC_RNaseH UvrC_RNaseH_dom HHH_5 CDP-OH_P_transf Bac_DNA_binding PBECR5

MBQ2175865.1
JAFPGB010000568.1:35-9696
Alphaproteobacteria bacterium

ART-PolyVal LPD23 PBECR5 ? ? Phage_tail_APC ?

MBQ2396426.1
JAFPAI010000046.1:19-17131
Bacteroidales bacterium

PBECR5 BppU_N ? ? Phage_holin_Dp1 Recombinase Resolvase ? ?

MBQ3221220.1
JAFQGG010000024.1:11998-31258
Clostridia bacterium

NMT1 tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C Ribosom_S12_S23 Ribosomal_S7 GTP_EFTU EF-G_D2 EFG_III EFG_IV EFG_C GTP_EFTU GTP_EFTU_D2 GTP_EFTU_D3 LPD3 ART-PolyVal PBECR5 Rce1-like SBP_bac_8 BPD_transp_1 Plasmid_mob_TrbF BPD_transp_1 TOBE_2 ABC_tran Cupin_2 HTH_3 Ribosomal_S20p Asp_Glu_race

MBQ3379008.1
JAFQZS010000021.1:172-3447
Clostridia bacterium

ParB_N PBECR5 MuF_C

MBQ3431580.1
JAFQYR010000095.1:1-12257
Clostridia bacterium

Phage_capsid_4 ? ? ? ? ? ? ART-PolyVal ART-PolyVal PBECR5

MBQ3634192.1
JAFPZF010000007.1:1-9777
Bacteroidales bacterium

PBECR5 MuF_C ? Amidase_3 DUF8434 MlaD Helical_needle Peptidase_M23 LysM S1 S1 S1 S1 S1 S1 STAS Anti-Pycsar_Apyc1 Lactamase_B_2

MBQ3980967.1
JAFPMX010000209.1:1-1624
Treponema sp.

PBECR5 SHOCT

MBQ4184110.1
JAFPNS010000144.1:1-5060
Bacteroidales bacterium

MuF_C ART-PolyVal PBECR5 ART-PolyVal ART-PolyVal ART-PolyVal

MBQ4614091.1
JAFQGD010000031.1:33498-55506
Akkermansia sp.

? AIRS_C AIRS CstA CstA Transgly Transpeptidase ? Aminotran_5 LPD3 ART-PolyVal PBECR5 Discoidin ? Caps_synth ? Aminotran_5 DUF4131 Competence ?

MBQ5796294.1
JAFNZU010000651.1:1-4606
Kiritimatiellae bacterium

ddrB-ParB PBECR5

MBQ6048688.1
JAFRTO010000164.1:1-5810
Oscillospiraceae bacterium

MuF_C PBECR5

MBQ6217334.1
JAFRQF010000026.1:2284-26086
Erysipelotrichaceae bacterium

RVT_1 ? Phage_XkdX ? CHAP ? ? MuF_C PBECR5 LPD38 ? HSP90 HSP90 HATPase_c_3 YbeY DAGK_prokar dCMP_cyt_deam_1 MMR_HSR1 KH_2 RecO_N RecO_C

MBQ6398195.1
JAFRGS010000107.1:1-13365
Oscillospiraceae bacterium

? PBECR5 ? ? Phage_capsid_4 ? ? ? ?

MBQ68923.1
PBTC01000014.1:51662-66971
Candidatus Saccharibacteria bacterium

YdaS_toxin PcfJ ABC-ATPase AAA_23 ? PcfJ ? ? PBECR5 ? ? ? UDG ? ? ?

MBQ7247132.1
JAFSTX010000033.1:37487-63157
Lachnospiraceae bacterium

? Phage_capsid_3 ? ? ? ? ? RadC PBECR5 ? ? Collagen LbR_Ice_bind Ribosomal_L32p ? Zn_ribbon_Top1 Zn_ribbon_Top1 Topoisom_bac Toprim WHD_DprA DNA_processg_A

MBQ7249371.1
JAFSTW010000235.1:1-2117
Deltaproteobacteria bacterium

RecA_N PBECR5

MBQ7534684.1
JAFSYO010000068.1:1-3210
Stomatobaculum sp.

LPD3 PBECR5

MBQ7748653.1
JAFSSD010000283.1:1-2759
Paludibacteraceae bacterium

PBECR5 Kinase-PolyVal

MBQ8044103.1
JAFUIH010000269.1:8222-30478
Clostridia bacterium

? ? ? ? ? ? ? ART-PolyVal PBECR5 DUF5021 ? ? Pect-lyase_RHGA_epim ? Holin_SPP1 Amidase_2 PG_binding_1

MBQ8603913.1
JAFUNP010000016.1:53098-72934
Oscillospiraceae bacterium

Glyco_hydro_25 Cnn_1N ? ? ? ? ? ? PBECR5 MuF_C ? DUF1003 Pyrophosphatase UPF0122 SRP54_N SRP54 SRP_SPB Ribosomal_S16 KH_KhpA-B

MBQ8637161.1
JAFUOC010000006.1:40823-67147
Clostridia bacterium

? ? NLPC_P60 ? Phage_holin_6_1 NLPC_P60 ? MutS_I PBECR5 EFP_N EFP Elong-fact-P_C YnfU HAD_2 PcrA_UvrD_tudor UvrD_C UvrD-helicase PDZ_2 Peptidase_S55 Cation_ATPase_N E1-E2_ATPase Cation_ATPase Hydrolase_3 Cation_ATPase_C SDF

MBQ8637473.1
JAFUOC010000021.1:33757-60081
Clostridia bacterium

? ? NLPC_P60 ? Phage_holin_6_1 NLPC_P60 ? MutS_I PBECR5 EFP_N EFP Elong-fact-P_C YnfU HAD_2 PcrA_UvrD_tudor UvrD_C UvrD-helicase PDZ_2 Peptidase_S55 Cation_ATPase_N E1-E2_ATPase Cation_ATPase Hydrolase_3 Cation_ATPase_C SDF

MBQ8796534.1
JAFTWB010000028.1:228-19246
Clostridia bacterium

? ? ? Calcineurin ? ? ? LPD3 PBECR5 ? ? Glyco_hydro_25 Dockerin_1 Tad2-like Holin_SPP1

MBQ9163138.1
JAFUEC010000001.1:12-12521
Bacteroidaceae bacterium

Methyltransf_23 PBECR5 ? DUF4421 Ribosomal_S16 HD_assoc_2 HD OMPdecase RF-1 PCRF AIRS_C AIRS

MBR1566046.1
JAFUWU010000178.1:1-2639
Oscillospiraceae bacterium

PBECR5 ART-PolyVal Methyltransf_23

MBR1654829.1
JAFUYR010000003.1:205-32168
Prevotella sp.

Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase YobJ ? ? ? InPase LPD1 ART-PolyVal PBECR5 LPD3 DUF8192 P22_portal ? SU10_MCP ? ? ?

MBR1789916.1
JAFVBD010000099.1:1-4086
Bacteroidaceae bacterium

PBECR5 ART-PolyVal

MBR1885833.1
JAFVDB010000163.1:198-6467
Schwartzia sp. (in: firmicutes)

PBECR5 LPD23 SprT-like ? SLH

MBR1918110.1
JAFVDU010000007.1:875-2754
Spirochaetales bacterium

? PBECR5

MBR2145062.1
JAFVMS010000386.1:1-2462
Akkermansia sp.

PBECR5

MBR3635910.1
JAFXOG010000297.1:1-15612
Lachnospiraceae bacterium

? ? ? ? ? ? ? MuF_C ART-PolyVal PBECR5 ART-PolyVal ART-PolyVal ART-PolyVal ART-PolyVal

MBR3903251.1
JAFWYD010000061.1:45754-75022
Akkermansia sp.

T7_gp14 Head-tail_con ? ? ? ? ? MuF_C PBECR5 Phage_integrase Phage_int_SAM_5 Amidase_3 Amidase_3 NDK AMP_N Peptidase_M24 MauE ?

MBR3919098.1
JAFWZH010000235.1:1-28715
Clostridia bacterium

DUF4368 ? ? ? DUF859 ? PBECR5 ART-PolyVal LPD3 ART-PolyVal Collagen Gp10_C ? Phage_holin_5_2 PG_binding_1 Lipase_GDSL_2 CE2_N Glyco_hydro_28 Pect-lyase_RHGA_epim ?

MBR3973253.1
JAFXAV010000036.1:63590-81809
Oscillospiraceae bacterium

DUF739 ? ? ? IstB_IS21 PDDEXK_15 ? PBECR5 DUF2730 HTH_26 ? ? ? RepB-RCR_reg ? ?

MBR4029355.1
JAFWXI010000046.1:17141-47276
Alistipes sp.

? ? HAD_PNKP ? ? ? NOT1_connector ART-PolyVal ART-PolyVal PBECR5 LPD23 P22_portal ? SSB ? ? ? SU10_MCP

MBR4310754.1
JAFYMO010000526.1:16721-39987
Akkermansia sp.

? ? Phage_capsid_2 ? ? ? ? LPD3 PBECR5 MuF_C Phage_integrase

MBR4457580.1
JAFYGW010000009.1:148-21126
Clostridia bacterium

? ? Phage_capsid_4 ? ? ? ? ART-PolyVal ART-PolyVal PBECR5 ZMYND8_CC

MBR5309513.1
JAFYOZ010000003.1:1-5649
Oscillospiraceae bacterium

Amidase_3 SPOR ? P22_portal ? PBECR5

MBR5895587.1
JAFZHC010000058.1:13851-43160
Akkermansia sp.

? ? ? ? ? ? Glucosaminidase MuF_C PBECR5 Phage_integrase His_biosynth SUFBD_core SUFBD_core SufBD_N ABC_tran FUR ArgJ

MBR6049622.1
JAFXWT010000007.1:1-2094
Clostridia bacterium

PBECR5

MBR6530760.1
JAFYLE010000134.1:1-4887
Clostridia bacterium

PBECR5 ART-PolyVal

MBR6709563.1
JAFXHZ010000361.1:1-2576
Clostridia bacterium

PBECR5

MBS0507026.1
JAFEES010000093.1:1-9025
Proteobacteria bacterium

SQS_PSY BSH_RND Beta-barrel_RND_2 YknX_C ACR_tran ? PBECR5 ? BrnT_toxin BrnA_antitoxin Trigger_N

MBS5334155.1
JAGZGF010000004.1:14833-36356
Bacillota bacterium

? ? ? ? SU10_adaptor ? ? LPD3 PBECR5 LPD3 ? DUF859 ? Phage_holin_Dp1 DUF948 ? Amidase_2 PG_binding_1

MBS5575336.1
JAGZCH010000009.1:42389-58111
Bacillota bacterium

? ? ? ? Phage_gp49_66 CARD DUF6273 PBECR5 DnaJ HTH_3 HTH_38 rve Pribosyltran ? Acetylase ATC_hydrolase

MBS6108622.1
JAHAGO010000024.1:1448-20650

Subdoligranulum variabile

Phage_capsid_4 ? ? ? ? ? ? PBECR3 PBECR5 ? Collagen ? Phage_XkdX ? Gp10_C ?

MBS6175273.1
JAHAFJ010000022.1:242-18225
Clostridiales bacterium

? ? ? Gp58 Peptidase_S74 DUF3322 ? SHOCT_2 PBECR5 ? ? ? ? Amidase_2

MBS6372342.1
JAHACG010000043.1:4021-16784
Oscillospiraceae bacterium

Ribosomal_L1 Ribosomal_L11 Ribosomal_L11_N KOW NusG SecE Ribosomal_L33 HTH_3 ? eCIS_core PBECR5 ? Mac Hexapep GATase_6 SIS SIS ? ? Glt_symporter MatE MatE

MBS6372713.1
JAHACG010000064.1:187-10285
Oscillospiraceae bacterium

ADPrib_exo_Tox PBECR5 ART-PolyVal ? ? ? ? ? ? Amidase_2 SLH

MBS7087923.1
JAGZQS010000033.1:19697-37758
Klebsiella aerogenes

GP7 Bbp16 ? ? Phage_T7_Gp13 T7_gp14 NLPC_P60 DUF6861 MuF_C PBECR5 PBECR5 ?

MBX3354535.1
JAHBWY010000002.1:42978-62668
Phycisphaeraceae bacterium

MGS AICARFT_IMPCHas 3D MltA TatA_B_E DUF2868 MMR_HSR1 DUF3482 Patatin Dockerin_1 Peptidase_M23 PBECR5 TM2 Oxidored_FMN MFS_3 MscL MgtE_N CBS CBS MgtE Abhydrolase_1 AhpC-TSA

MCA9244511.1
JAGQOO010000163.1:44-7500
Phycisphaerales bacterium

PBECR5 ?

MCC8176186.1
JAJBTE010000124.1:1-5429
Bacteroidales bacterium

PBECR5 LPD3 LPD23

MCD7797157.1
JAJQGZ010000152.1:68-3279
Clostridiales bacterium

fn3 Ribosomal_L27 Peptidase_Prp Ribosomal_L21p ? ? MuF_C PBECR5

MCD8282410.1
JAJPZE010000145.1:1-4738
Prevotella sp.

PBECR3 LPD23 LPD3 PBECR5

MCH4182979.1
JAKVKK010000031.1:8061-29845
Prevotella sp.

? Collagen Collagen ? ? NAGPA Phage_r1t_holin ? ComR_TPR LPD3 PBECR5 Phage_integrase Peptidase_S24 HTH_3 DUF739 ? MerR_1 ? ?

MCI5836228.1
JALELM010000001.1:672-22478
Veillonellaceae bacterium

Bbp16 ? ? Phage_nozzle Phage_T7_Gp13 ? SLT DUF6861 PBECR5 LPD3 Phage_T7_tail ? ? Phage_holin_4_1 Amidase_2 ? DUF3450

MCI6671844.1
JALFSW010000051.1:1-24971
Prevotella sp.

? ? T7_gp14 ? ddrB-ParB InPase PBECR5 LPD3 MuF_C P22_portal ? ? ? ? SU10_MCP ?

MCI6948271.1
JALFEM010000040.1:26826-49846
Oribacterium sp.

Peptidase_S74 Peptidase_S74 Peptidase_S74 ? DUF3597 PBECR3 PBECR5 Mobilization_B Phage_holin_5_2 ELLD Glyco_hydro_25 Choline_bind_3 ABC_tran OB_MalK TOBE_2 GGDEF_2 HTH_30 FtsX_ECD FtsX CCDC39 CC_PcsB Peptidase_M23

MCI7685420.1
JALFWT010000003.1:73781-86316
Clostridiales bacterium

CRISPR_Cas2 ? ? ? ? ? ? LPD3 PBECR5 ? DUF3293 Pyridox_ox_2 Peptidase_M41 AAA_lid_3 AAA FtsH_ext MFS_1 Acyl-ACP_TE_C Acyl-ACP_TE PP-binding

MCI7699566.1
JALFWB010000050.1:1-17010
Akkermansia sp.

? BATS Radical_SAM SBP_bac_3 SBP_bac_3 BPD_transp_1 PBECR5 ART-PolyVal SseB ? Ribosom_S12_S23 Ribosomal_S7 GTP_EFTU GTP_EFTU_D2 EFG_III EFG_IV EFG_C Ribosomal_S10 Ribosomal_L3

MCK9431609.1
JALNZO010000004.1:17237-41363
Candidatus Omnitrophota bacterium

? Transglutaminase ? ? ? ? ? ParB_N PBECR5 ? ? ? ? ? ?

MCL1829749.1
WRJA01000113.1:1-3120
Oscillospiraceae bacterium

PBECR5

MCL2182061.1
WQRV01000006.1:1-15023
Chitinispirillia bacterium

? SU10_adaptor LamininG ? ? ? ? PBECR5

MCM1170205.1
JAMPFX010000122.1:1-4965
Bacteroides sp.

LPD3 ART-PolyVal PBECR5 LPD23 DUF6782

MCM1218215.1
JAMPGK010000061.1:48-18451
Lachnospiraceae bacterium

? ? LPD3 PBECR5 ART-PolyVal ART-PolyVal ART-PolyVal ART-PolyVal ? Flg_new Zn_ribbon_TiaS ? ? ? Phage_capsid_4

MCM1455467.1
JAMPBR010000021.1:1-13737
Bacteroides sp.

? ? ? GP3_package ? SU10_portal Hint_2 PBECR3 PBECR5 ART-PolyVal Methyltransf_23 ART-PolyVal

MCM1555795.1
JAMOZR010000026.1:1-8670
Bacteroides sp.

? PBECR1 LPD3 ART-PolyVal PBECR5

MCP3681485.1
JAAGZO010000152.1:9123-15124
bacterium

? ? ? HTH_38 NAT10_TcmA_helicase SU10_portal ? PBECR5

MCP4493229.1
JAAEPM010005132.1:1-23125
Gammaproteobacteria bacterium

? LamininG ? ? ? Adhesin_P1_N ? PBECR5 ART-PolyVal LPD38 ? ? ? ? ? ? ?

MCP5058977.1
JAAETT010000462.1:1-4736
bacterium

?

MCP5230922.1
JACKMD010000034.1:21231-41475

Zoogloeaceae bacterium

? E1_FCCH ? ? ? ? ? PBECR5 MuF_C ART-PolyVal ? ? ? ? Acb2_Tad1_hairpin ? ?

MCP5244887.1
JACKLS010000006.1:43615-63372
Burkholderiales bacterium

? ? ? DUF2824 ? ? ? ddrB-ParB PBECR5 ART-PolyVal ART-PolyVal LPD38 ? ? ? ? ? Zn_ribbon_recom Recombinase Resolvase ?

MCP5246148.1
JACKLS010000033.1:1-14890
Burkholderiales bacterium

? ? ? Acetyltransf_4 ? ? DUF5398 ddrB-ParB LPD23 PBECR5 ART-PolyVal Peptidase_M30

MCQ2220068.1
JAKSLK010000007.1:4739-25525
Prevotella sp.

NAGidase Phage_capsid_3 ? ? ? ? DUF6384 ART-PolyVal ART-PolyVal PBECR5 Peptidase_M23 Phage_holin_1 Amidase_2 PG_binding_1 ? ? ? ? DUF2730

MCR5566208.1
JAILNB010000055.1:23056-48006
Clostridiales bacterium

Flg_new_2 ? ? ? ? Flg_new_2 Flg_new_2 ? ART-PolyVal PBECR5 ? ? NLPC_P60 PG_binding_1 DUF8429 Phage_holin_1

MCS3211691.1
JANUOK010000006.1:2275586-2288397
Bacteroides thetaiotaomicron

? ParA-Soj-NTPase UBA MethyltransfD12 ? ? ? PBECR5 ? ? TrwB_AAD_bind ? ? Amidoligase_2 ?

MCX4257908.1
JAPKXL010000057.1:1-8231
Oscillospiraceae bacterium

? PBECR5

MDC0088173.1
JAOLWU010000003.1:33489-53592
Akkermansiaceae bacterium

DUF1353 ? ? ? TMEM126 T7_gp14 ? PBECR5 ? ? Amidase_2 ? ? DUF3244 ?

MDC7250459.1
JAQQAG010000159.1:6-14399
Pleomorphochaeta sp.

IstB_IS21 ? ? ? HTH_23 ATP-synt_ab Terminase_6C ? LPD3 ART-PolyVal PBECR5 PBECR5 ?

MDD3073606.1
JAQVAX010000046.1:1-10149
Eubacteriales bacterium

? Gly-zipper_Omp ? ddrB-ParB ART-PolyVal PBECR5

MDD3535408.1
JAQVIX010000025.1:3072-31064
Candidatus Cloacimonadota bacterium

ABC_tran SNase ? HKD SNF2-rel_dom Helicase_C MmeI_Mtase Eco57I M_Eco57I_C ? tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C ddrB-ParB MuF_C ART-PolyVal PBECR5 ART-PolyVal ART-PolyVal Amidase_2

MDD3884562.1
JAQVVH010000002.1:20948-42338
Gallionella sp.

GP7 Bbp16 ? ? Phage_nozzle ? ? LPD22 PBECR5 End_tail_spike Peptidase_S74 FGS bAvd RVT_1 Phage_holin_2_3 Phage_lysozyme ?

MDD4012456.1
JAQVSP010000176.1:1-3337
Sphaerochaetaceae bacterium

? DUF2026 PBECR5

MDD4292097.1
JAQVMT010000026.1:615-15519
Clostridia bacterium

GalKase_gal_bdg GHMP_kinases_N PadR HAAS DUF4097 DUF4097 PBECR5 LRRs Flg_new ? ? ? DUF4304 ABC_tran ABC_membrane ABC_tran ABC_membrane

MDD4381719.1
JAQWFF010000001.1:1-13221
Candidatus Dojkabacteria bacterium

? ? ? ? ? ? LysM PBECR5

MDD4913285.1
JAQTUV010000009.1:140-18671
Sideroxydans sp.

GP7 Bbp16 ? ? ? ? ? LPD22 ART-PolyVal ART-PolyVal PBECR5 Pect-lyase_RHGA_epim Beta_helix ? ? Tape_meas_lam_C ? ? ?

MDD5910626.1
JAQXOG010000008.1:645-18384
Bacteroidales bacterium

DUF6808 NUDIX HD_3 NTP_transf_2 Methyltransf_23 LPD3 ART-PolyVal PBECR5 Kinase-PolyVal PBECR2 ART-PolyVal Nuc_deoxyri_tr2 SIMPL FlK EamA EamA Glyco_hydro_2_N2 Glyco_hydro_2 Glyco_hydro_2_C Ig_GlcNase Porin_O_P

MDD6363068.1
JAQYBH010000109.1:1-12713
Lachnospiraceae bacterium

Phage_Mu_F ? ? ? MuF_C PBECR5 ? OMPdecase Pribosyltran DUF6142 Peptidase_M29 AIRC AIRS AIRS_C

MDD7288223.1
JAQYFK010000017.1:27-6910
Clostridiales bacterium

DUF4097 HTH_18 Peptidase_M20 ADIP LPD3 PBECR5 DUF1202

MDE5944559.1
JAIDFC010000102.1:2773-29290
Rikenella sp.

DUF4272 HTH_3 ? ? T7_gp14 ? LPD14 PBECR5 MuF_C LPD3 LPD38 Fic P22_portal GmrSD_N AAA_15 AAA_15 DUF3696 ? ?

MDE6574146.1
JAIDWX010000106.1:1-3163
Muribaculaceae bacterium

PBECR5 LPD3

MDE8717183.1
JARFIF010000008.1:61728-85559
[Eubacterium] siraeum

P22_portal Peptidase_Prp ? Phage_capsid_3 ? ? ? LPD3 PBECR5 Fe-ADH Zeta_toxin ? Fumerase_C ? IreB Polysacc_deac_1

MDO4389233.1
JAUNFC010000057.1:1-14866
Eubacteriales bacterium

Glyco_tranf_2_3 Arginosuc_synth Arginosuc_syn_C SBP_bac_3 BPD_transp_1 OTCace_N OTCace CPSase_sm_chain GATase CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS DAP_epimerase DAP_epimerase MuF_C PBECR5

MDO4494945.1
JAUNCW010000005.1:7339-27920
Clostridiaceae bacterium

? ? ? DUF874 ? ? ? ART-PolyVal PBECR5 ? ? ? ? Phage_holin_1 CHAP ? CotJB

MDO5035841.1
JAUNLH010000011.1:187-21832
Porphyromonas sp.

Transglutaminase Big Sigma54_AID Sigma54_CBD Sigma54_DBD GCV_H AIRC Hydantoinase_A ? LPD3 ART-PolyVal PBECR5 MuF_C ? Toluene_X Peptidase_M60 Discoidin BACON MFS_1 MMR_HSR1 YegS_C DAGK_cat MPI_cupin_dom PMI_typeI_cat

MDO5398925.1
JAUNPN010000163.1:1-6330
bacterium

PBECR5

MDP1931891.1
JAUXRM010000105.1:22511-43677
Gammaproteobacteria bacterium

? ? ? ? ? ? ? PBECR5 MuF_C ART-PolyVal ART-PolyVal LPD38 ? ? ? DUF4953 NadA TPR_CcmH_CycH Fis1_TPR_C Peptidase_M48 TusA

MDP9135692.1
JALYJO010000740.1:1-2316
Actinomycetota bacterium

PBECR5

MDR0535825.1
JAIRPL010000047.1:113613-141365
Puniceicoccales bacterium

PMT_2 GDP_Man_Dehyd Met_synt_B12 Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans PA14 Fer4_20 Pyr_redox_2 Pyr_redox_2 UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C Amidohydro_1 PBECR3 ART-PolyVal PBECR5 ART-PolyVal ART-PolyVal ART-PolyVal PBECR5 LPD3 PBECR5 ? ? Hydrolase ? GFO_IDH_MocA GFO_IDH_MocA_C3 TatA_B_E ABC-3

MDR0657782.1
JAIRNG010000034.1:1-2677
Mediterranea sp.

DUF5712 ART-PolyVal PBECR5

MDR2356883.1
JAITKI010000033.1:20106-39596
Oscillospiraceae bacterium

Helicase_C ResIII ? BrxC_BrxD BrxC_BrxD BrxC_BrxD Receiver His-Kinase His-Kinase Receiver His-Kinase His-Kinase HAMP dCache_1 DDE_Tnp_1 ? MuF_C ART-PolyVal LPD3 PBECR5

MDR2513855.1
JAITGM010000010.1:31272-52767
Christensenellaceae bacterium

Phage_capsid_4 ? ? ? ? DUF859 PG_binding_1 PBECR3 PBECR5 Collagen Collagen Collagen Collagen Collagen Collagen Beta-prop_FBX42 Kelch_KLHDC2_KLHL20_DRC7 SH3_3 ? ? ? Ribosomal_L19

MDR2671376.1
JAITDM010000246.1:1-18417
Oscillospiraceae bacterium

Collagen Collagen ? ? DUF6273 ? ? ? PBECR5 ? NAGPA Peptidase_M15_3

MDV7391591.1
JAWMAM010001771.1:1-540
Arthrospira platensis SPKY1

PBECR5

MDY3109109.1
JAXFYA010000073.1:1-2071
Lachnospiraceae bacterium

PBECR5

MDY5350550.1
JAXFIC010000168.1:4303-24875
Candidatus Ventricola sp.

Peptidase_Prp ? Phage_capsid_3 ? ? ? PG_binding_1 ART-PolyVal PBECR5 Amidase_2 SPOR SH3_3 ? DUF1686 ? ? ? ?

MEA2004377.1
JAYEIW010000260.1:1-1756
archaeon

PBECR5

MEA3463273.1
JAYEOJ010000679.1:1-1615
Bacteroidota bacterium

PBECR5

MEA4934810.1
JAYFPV010000487.1:1078-19318
Lawsonibacter sp.

? ? ? ? ? ? ? MuF_C PBECR5 ART-PolyVal Phage_holin_4_1 ? Amidase_2 ?

MEE0527080.1
JAXIYU010000085.1:79-16298
Prevotellamassilia sp.

Phage_int_SAM_5 ? ? LPD14 ddrB-ParB PBECR2 InPase LPD3 ART-PolyVal PBECR5 LPD23

MEE3458247.1
JAZASE010000002.1:708-17037
Faecousia sp.

ART-PolyVal PBECR5 Filament ? Phage_capsid_4 ? ? Y_phosphatase3 Phage_r1t_holin Amidase_2 SPOR

MEE4109481.1
JAZBNX010000442.1:6455-21000
Halieaceae bacterium

RadC PBECR5 LPD38 LemA DUF4156 ? Phage_integrase Peptidase_S58 ? Tricorn_N Tricorn_N Peptidase_S9

MFS1429135.1
JAJGZM020000276.1:411-6658
Vibrio splendidus

? ? ? NDK Band-7 ? ?

NBI43228.1
JAAAXK010000028.1:289-22728
[Haemophilus] felis

? Terminase_2 Terminase_6N SU10_portal Pribosyltran PBECR1 PBECR5 LPD3 ART-PolyVal MuF_C LPD38 ? ? Phage_capsid_4 ? ? ? Phage_fiber_2 Phage_fiber_2 Collar PTR

NCD04937.1
RZYO01000009.1:12915-37361
Spirochaetia bacterium

Phage_rep_org_N IstB_IS21 ? NUMOD1 HTH_23 Terminase_6C ? ART-PolyVal PBECR5 Phage_integrase ? Na_H_Exchanger PTS_EIIA_2 Glyco_hydro_125 LacI Peripla_BP_3 SBP_bac_1

NDF13186.1
RGZB01000070.1:84-6906
Proteobacteria bacterium

ART-PolyVal PBECR5 ?

NLC97438.1
JAAZCQ010000057.1:1-3405
Erysipelotrichaceae bacterium

PBECR5

NLV83303.1
JAAYDK010000061.1:1-1548
Spirochaetales bacterium

ART-PolyVal PBECR5

NTU49312.1
JAAXWB010000001.1:19295-38649
Desulfobulbaceae bacterium

Phage_stabilise Gp80 DUF2807 ? ? ? ? SLT ART-PolyVal PBECR5 ? Gp10_C ? VanY Peptidase_M15_4 DUF6905 ? ?

NYT44606.1
JACCEQ010000003.1:253820-273791
Alcaligenaceae bacterium

Phage_T7_Capsid GP7 Bbp16 ? ? T7_gp14 SLT LPD22 ART-PolyVal PBECR5 ? ? ? ? ? ? Arc

NYT45145.1
JACCEQ010000004.1:390257-408166
Alcaligenaceae bacterium

? GP7 Bbp16 ? ? T7_gp14 SLT LPD22 ART-PolyVal ART-PolyVal PBECR5 ?

OLA00147.1
MNRF01000336.1:38141-56544

Clostridiales bacterium 42_27

? Phage_holin_4_1 Peptidase_M15_3 ? ? CCDC39 ? eCIS_core PBECR5 ? ? ? ? BclA_C DUF7841 GerA

PIV83341.1
PFFB01000066.1:1-6078

Candidatus Moranbacteria bacterium CG17_big_fil_post_rev_8_21_14_2_50_44_12

? PBECR5 ?

PNC26046.1
PJKD01000001.1:146415-181383
Akkermansia muciniphila

? HTH_17 HTH_17 ? ? ? ? PBECR5 ART-PolyVal ART-PolyVal ART-PolyVal ART-PolyVal ART-PolyVal PBECR5 ART-PolyVal TIR TIR Abi_2 KAP_NTPase KAP_NTPase TEN_YD-shell TEN_YD-shell TEN_YD-shell TEN_YD-shell ? Glyco_transf_25

QDP63586.1
MK892787.1:15667-46144
Prokaryotic dsDNA virus sp.

? ? ? ? ? ? ? ART-PolyVal PBECR5 ? ? HTH_26 Helicase_C ? AAA_24 ?

QHV66237.1
CP029703.1:2236152-2262864
Akkermansia muciniphila

? ? Phage_capsid_2 ? ? ? ? LPD3 ART-PolyVal PBECR5 ? Ligase_CoA CoA_binding Ligase_CoA ATP-grasp_2 Histidinol_dh ? ? TPRs TPRs TPRs

RHQ17418.1
QUFC01000008.1:1-20294
Lachnospiraceae bacterium AM48-27BH

? Phage_XkdX ? ? Gp58 Peptidase_S74 DUF3597 ? LPD3 PBECR5 Phage_holin_4_1 DUF4406 DUF6275 Tad2-like Tad2-like Amidase_2 NLPC_P60 Syndecan LytR_cpsA_psr

RHU37239.1
QUKE01000008.1:115695-132840
Clostridium sp. TM06-18

? LemA ? ? Wzy_C DUF2976 ? PBECR5 Relaxase HTH_26 DNA_methylase Eco57I ? DUF6414 AAA_2 ?

RHV86968.1
QUME01000030.1:4778-24445
Lachnospiraceae bacterium OF09-33XD

? ? HTH_3 Gp49 ? DUF4473 ? MuF_C PBECR5 MuF_C ? BppU_N ? ? BppU_N Phage_holin_4_1

RKW31321.1
RBKB01002098.1:1-2230
Lachnospiraceae bacterium

MuF_C PBECR5

RKW44796.1
RBKG01000273.1:1-2447
Porphyromonas sp.

PBECR5 MuF_C

TXH51590.1
SSEM01000278.1:1-9533
Desulfurellales bacterium

PGDYG LPD23 Zeta_toxin PBECR5

UWI04662.1
OP075507.1:31647-42911
Bacteriophage sp.

Phage_holin_6_1 ? Amidase_2 SH3_3 TM2 DUF4236 ? HTH_3 HTH_38 rve PBECR5 Collagen CBM_5_12

UYJ12555.1
CP107211.1:1638884-1658813
Oscillospiraceae bacterium

? ? ? ? ? ? PBECR3 LPD3 PBECR5 ? DUF6273 bZIP_1 CD1375-like Glyco_hydro_25 ? Holin_SPP1

WP_229092123.1
NZ_JADNAF010000059.1:1302-27649
Bacteroides ovatus

? MFS_1 MFS_1 Lactamase_B_2 Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase DUF5681 T7_gp14 ? LPD14 InPase ART-PolyVal PBECR5 ART-PolyVal P22_portal ? ? Rhomboid ? SYF2 SU10_MCP

WP_138307309.1
NZ_SPHS01000004.1:84338-108428
Flavonifractor plautii

Tad2-like ? Amidase_2 Gp58 ? ? ? MuF_C LPD23 PBECR5 ? HTH_3 HTH_38 rve Phage_integrase Phage_int_SAM_3 Phage_integrase Phage_int_SAM_3 ? DUF6809

WP_118726646.1
NZ_QUIK01000013.1:76592-101914
Clostridium sp. AM42-4

DUF6275 VanY Peptidase_M15_4 Choline_bind_3 Choline_bind_3 ? ? Peptidase_S74 Peptidase_S74 ? PBECR3 PBECR5 His-Kinase His-Kinase Receiver ParB_N DUF7768 DNA_methylase DNA_methylase ArdA ? DUF3801

WP_147590284.1
NZ_CP042993.1:402404-419720
Histophilus somni

Phage_holin_2_3 Phage_lysozyme DUF2570 LysC ? Terminase_6N SU10_portal SU10_portal PBECR5 LPD3 LPD3 LPD38 HTH_3 ? Herpes_UL25 Phage_capsid_4 SAP

WP_113062627.1
NZ_UATH01000001.1:1606853-1623346
Oligella urethralis

Cu-oxidase Cu-oxidase_3 HicB_lk_antitox HicA_toxin IMPDH GATase NAD_synthase GMP_synt_C MuF_C Transposase_mut PBECR5 ? MuF_C ArsB Dus DUF1653 MamF_MmsF ATP-synt_ab Rho_RNA_bind Rho_N

WP_018865146.1
NZ_KB897952.1:45718-72629
Thioalkalivibrio sp. ARh3

? ? ? ? ? Gly-zipper_YMGG ? ART-PolyVal PBECR5 ART-PolyVal ART-PolyVal ? ? GntP_permease Gly_kinase SBP_bac_3 PAS GGDEF EAL EAL GGDEF PAS Receiver His-Kinase HDOD

WP_021773540.1
NZ_KI273102.1:43095-63037
Oribacterium sp. oral taxon 078 str. F0263

? ? Peptidase_S74 Gp58 Peptidase_S74 Peptidase_S74 ? DUF3597 PBECR5 ? DUF4406 DUF6275 Phage_holin_5_2 ? ELLD ?

WP_026651320.1
NZ_JHWL01000003.1:1-19038
Butyrivibrio proteoclasticus P6B7

DUF6273 RVT_1 ? CD1375-like ? ? ? ART-PolyVal PBECR5 BppU_N ? ? Phage_holin_6_1 Glyco_hydro_25 CW_7 ?

WP_046997313.1
NZ_JAIQ01000165.1:1-12535
Aliarcobacter butzleri L348

? Mor SU10_portal ? DUF2508 ? ? PBECR5 SU10_MCP SU10_MCP ? Phage_holin_4_1

WP_065236563.1
NZ_JTJR01000005.1:8355-26433
Gallibacterium genomosp. 3

Phage_holin_2_3 Phage_lysozyme DUF2570 LysC Phage_T4_gp2 ? SU10_portal PBECR1 PBECR5 MuF_C LPD38 ? Peptidase_S24 HTH_3 ? Bro-N Spatacsin_C ? ? Not3

WP_066054517.1
NZ_LWEE01000062.1:210947-226050
Halioglobus sp. HI00S01

? ? DNA_ligase_aden OB_DNA_ligase HHH_2 BRCT ? ? ? ? PBECR5 ? ? ? DnaT DnaT DnaT ? ? ?

WP_143039370.1
NZ_FNWD01000002.1:551223-576297
Millionella massiliensis

? ? ? ? ERF ? ? PBECR2 PBECR5 MuF_C LPD3 MuF_C LPD38 P22_portal ? ? ? ? ? ?

WP_102712570.1
NZ_PJKA01000006.1:102782-134373
Akkermansia muciniphila

? ? MarR_2 Terminase_2 ? ? ? ART-PolyVal ART-PolyVal PBECR5 ART-PolyVal ART-PolyVal Methyltransf_23 ? Zeta_toxin ? ? ? ? KAP_NTPase

WP_107683993.1
NZ_PZKL01000037.1:120891-138976
Aeromonas veronii

? zf-dskA_traR RlaP ALP_N MreB-like_C ? ? ? PBECR5 ? UvrD_C UvrD-helicase ? ? ? ? Receiver TPRs

WP_112147640.1
NZ_PRLE01000001.1:175754-197239
Faecalibacterium prausnitzii

Phage_capsid_4 ? ? ? ? ? ? MuF_C PBECR5 MazE_antitoxin YafQ_toxin ? ? ? ? ?

WP_282193767.1
NZ_CAOJGF010000028.1:8462-39351
Muribaculum intestinale

Pept_tRNA_hydro Ribosomal_TL5_C Ribosomal_L25p Ribosomal_L25p ? Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase ? T7_gp14 ? Kinase-PolyVal InPase PBECR5 LPD3 MuF_C LPD38 P22_portal ? ? Rhomboid ? SU10_MCP ?

WP_320226904.1
NZ_JAVIJC010000014.1:59964-79644
Mesorhizobium captivum

? Gp10-like ? GH85_C Phage_nozzle Phage_T7_Gp13 T7_gp14 VanY LPD22 PBECR5 ? ? DUF1515 ? ? ? DUF2116

WP_130837180.1
NZ_LR130576.1:277553-299407
Lachnoclostridium sp. Marseille-P6806

? ? ? ? ? Acyl_transf_3 Sel_put PBECR5 PBECR3 BppU_N ? ? Phage_holin_6_1 Phage_lysozyme SH3_3 Acyl_transf_3 Acyl_transf_3

WP_132301711.1
NZ_CP170642.1:1788541-1812102
Lonepinella koalarum

? ? ? ? ? Phage_lysozyme ? ddrB-ParB PBECR1 MuF_C ART-PolyVal PBECR5 LPD3 LPD38 ? ? ? ? ? ? FGAR-AT_N FGAR-AT_linker AIRS_C FGAR-AT_PurM_N-like AIRS_C GATase_5

WP_210822612.1
NZ_JAGPWB010000004.1:15400-36012
Hydrogenophaga aromaticivorans

DUF1353 ? ? ? ? ? ? PBECR5 ART-PolyVal LPD38 EamA EamA HSP15_C S4 His-Kinase His-Kinase HAMP_PhoQ 2CSK_N Trans_reg_C Receiver TctC TctB TctA

WP_200662353.1
NZ_JADGFW010000011.1:16342-39991
Psychrobacter sp. M9-54-1

DUF1376 IstB_IS21 AAA_22 IstB_IS21 DnaB DnaB_C Fe_dep_repress ? ? ? ddrB-ParB PBECR5 ART-PolyVal HTH_17 ? YdcA ? ? ? Excalibur SNase

WP_205806237.1
NZ_JAAGPI010000008.1:104429-133711
Burkholderia sp. Ac-20365

? Phage_integrase ? HTH_17 ? ? ? ddrB-ParB ddrB-ParB PBECR5 ? Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg Fil_haemagg TPS ESPR ? LysR_substrate HTH_1 Aldolase_II MFS_1 LysR_substrate HTH_1

WP_209215493.1
NZ_JAGHRG010000001.1:2947-15555
Sulfitobacter sp. R18_1

T4_Y16D Imp-YgjV ? ? ? ? ? PBECR5 Rdx ? ? ? ? ? ? ?

WP_246852507.1
NZ_JAFNBH010000013.1:153699-173800
Vibrio crassostreae

vWA AAA_lid_2 AAA_3 ? ? ERG4_ERG24 SPASM Radical_SAM Sulfatase_C Sulfatase GAF GerE PBECR5 ? ? ? ? His-Kinase DNA_gyraseB Toprim DNA_gyraseB_C DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C EAL GGDEF PAS

WP_216241884.1
NZ_JACVPG010000003.1:335878-354864
Polynucleobacter sp. Fuers-14

GP7 Bbp16 ? ? ? T7_gp14 ? LPD22 PBECR5 ? Mtd_N mtd_2nd bAvd RVT_1 ? LPP T4SS_pilin DUF7940

WP_235384105.1
NZ_JAKGBG010000064.1:13495-22881
Pseudoalteromonas sp. OFAV1

? ? ? PBECR5 SSB ? ? OMP_b-brl VWA ? DUF2989 DUF2989

WP_251453073.1
NZ_CALPCV010000043.1:78-10978
Parasutterella muris

PBECR5 LPD3 ABC-ATPase ? ? Gp53-like_C ? ?

WP_315569255.1
NZ_CAUUCS010000049.1:167-13256
Haemophilus parahaemolyticus

MuF_C PBECR1 MuF_C PBECR5 LPD3 LPD38 TPRs ? ? ? Bro-N ? ?

WP_270302288.1
NZ_JAQDDW010000001.1:244109-252517
Allobaileyella intestinalis

PBECR5 CRISPR_Cas2 Cas_Cas1 FtsH_ext AAA AAA_lid_3 Peptidase_M41 Pyridox_ox_2 ASCH ? PBECR5

WP_272962283.1
NZ_DDGO01000056.1:1-12138
Alcanivorax jadensis

? ART-PolyVal PBECR5 Phage_tail_beta Phage_tail_beta Beta_helix PAS ? Peptidase_M23 ? ?

WP_315387763.1
NZ_CAUVHT010000018.1:4125-30358
Hoylesella shahii

HTH_26 ? HTH_31 ? ? T7_gp14 ? ddrB-ParB RadC Kinase-PolyVal PBECR5 LPD38 P22_portal ? ? ? ? ? ?

WP_320892086.1
NZ_JAXEHQ010000003.1:124-7770
Dialister histaminiformans

PBECR5 ? PBECR5 Autotransporter SNO ThiG SOR_SNZ Amidohydro_1 AdoHcyase_NAD AdoHcyase

WP_276971532.1
NZ_JAQYDW010000013.1:1-6318
Eubacterium coprostanoligenes

FlgN PBECR5

WP_277029800.1
NZ_CAMVZA010000002.1:80599-101537
Turicimonas muris

? GP7 Bbp16 ? Phage_nozzle T7_gp14 ? ddrB-ParB PBECR5 Pox_H7 Collagen CBM_5_12 Collagen Collagen Sequestin Phage_tail_APC ? ?

WP_278474328.1
NZ_WHBV01000004.1:1-17712
Ruminococcus bromii

Phage_capsid_3 ? ? ? ? ? ? PBECR5 BppU_N

WP_278482363.1
NZ_JABZTH010000007.1:28545-59335
Hoylesella nanceiensis

PUA_4 Methyltrans_RNA ? PPDK_N PEP-utilizers PEP-utilizers_C Phage_int_SAM_5 ? T7_gp14 ? LPD13 InPase PBECR5 MuF_C LPD38 P22_portal ? ? Rhomboid ? ? SU10_MCP

WP_288080803.1
NZ_JALRGS010000321.1:1-12696
Pseudomonas sp.

Bbp16 ? ? Phage_nozzle Acetylase T7_gp14 ? LPD22 ART-PolyVal PBECR5

WP_288922475.1
NZ_CALKZQ010000038.1:11608-28923
uncultured Ottowia sp.

? NinB ? ddrB-ParB MuF_C MuF_C SocA_Panacea ? ? ? SocA_Panacea ? Terminase_3

WP_299145522.1
NZ_CABQJH010000011.1:18140-34380
uncultured Dialister sp.

RsgA_N RsgA_GTPase Ribul_P_3_epim HSP33 PNPase IF-2B AdoHcyase AdoHcyase_NAD Amidohydro_1 LPD3 LPD23 DUF6782 PBECR5 CLCA ? Phage_holin_4_1 ? Sequestin CCDC39 Phage_int_SAM_1 Phage_integrase tRNA_edit

WP_300726190.1
NZ_CASWJL010000008.1:6148-27789
uncultured Rikenella sp.

DUF418 LPD14 PBECR1 PBECR5 MuF_C LPD3 LPD38 Fic P22_portal ABC-ATPase ? ? ? ?

WP_303956349.1
NZ_JAQXIU010000007.1:14142-36440
[Clostridium] aminophilum

? Rubrerythrin ? ? ? ? DUF3597 DUF1542 PBECR5 Tropomyosin Phage_holin_5_2 Amidase_2 Choline_bind_3 GGDEF Hydrolase ?

WP_303991029.1
NZ_JAHAJN010000012.1:12059-32757
Negativicoccus succinicivorans

GP7 Bbp16 ? ? Phage_T7_Gp13 ? SLT DUF6861 MuF_C ART-PolyVal PBECR5 Phage_T7_tail LRRs

WP_304301738.1
NZ_JAFLKP010000277.1:6584-16856
Chromatium okenii

? ART-PolyVal ART-PolyVal PBECR5 ? ? LPD38 ? Terminase_2 Terminase_3 ?

WP_306557798.1
NZ_JAUURC010000021.1:33877-59085
Barnesiella intestinihominis

? SU10_MCP DUF5595 ? ? ? ? PBECR1 PBECR5 ART-PolyVal MuF_C LPD38 P22_portal ? PhdYeFM_antitox YoeB_toxin Phage_integrase Phage_int_SAM_5 Arm-DNA-bind_5

WP_323440845.1
NZ_JAYFRO010000061.1:41830-62164
Anaeromusa sp.

Bbp16 Rho_N ? ? Phage_T7_Gp13 T7_gp14 NLPC_P60 PBECR5 Lipase_GDSL_2 LamininG ? LamininG ? ? ?